

STAT 420/812

Computational Statistics

Sample Variance

$$(x_1, x_2, \dots, x_n) = c$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

$$\frac{\text{sum}(c - \text{mean}(c))^2}{\text{length}(c) - 1}$$

$$T \otimes T = T$$

$$T \otimes F = T$$

$$F \otimes T = T$$

$$T \otimes T = T$$

$$T \otimes T = T$$

$$F \otimes T = T$$

$$T \otimes T = T$$

$$T = 1, F = 0$$

$$Q = \#$$

$$P = \text{or}$$

	1	2	3	4	5	6	7	8	9	10
$c \geq 5$	F	F	F	F	F	T	T	T	T	T
$c < 10$	T	T	T	T	T	T	T	T	T	F
	F	F	F	F	F	T	T	T	T	F

matrix(1:20, 4, 5)



1	5	9	...	17
2	6	10		18
3	7	11		19
4	8	12	...	20

matrix(1:20, 4, 5, byrow = TRUE)

1	2	3	4	5
6	7	8	9	10
16	17	18	19	20

$$\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} * \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \end{bmatrix}$$

↓

$$= [1, 2, 3, 4, 5, 6] * [0.1, 0.2, 0.3]$$

$\begin{matrix} \times & \times & \times & \times & \times & \times \\ 0.1 & 0.2 & 0.3 & 0.1 & 0.2 & 0.3 \end{matrix}$

$$= [0.1 \quad 0.4 \quad 0.9 \quad 0.4 \quad 1.0 \quad 1.8]$$

$$= \begin{bmatrix} 0.1 & 0.4 \\ 0.4 & 1.0 \\ 0.9 & 1.8 \end{bmatrix}$$

T

name
peter

gender
m

Hw1
60

hw2
70

T

john

m

70

80

F

Lisa

f

70

80

↑

↑

strings

factor

↑

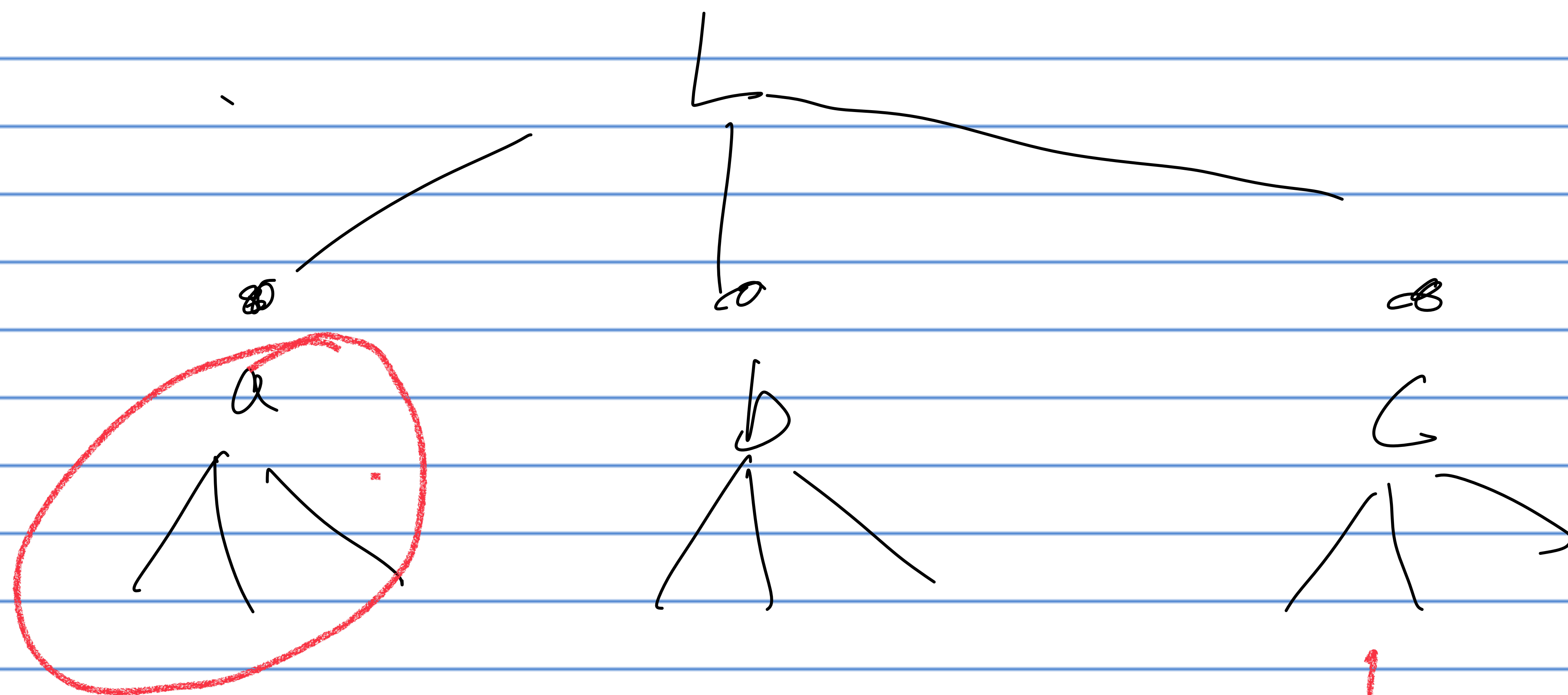
~~grades~~ gender == "m"

↑

row.names(A) ← . . .

col.names(A) ←

1

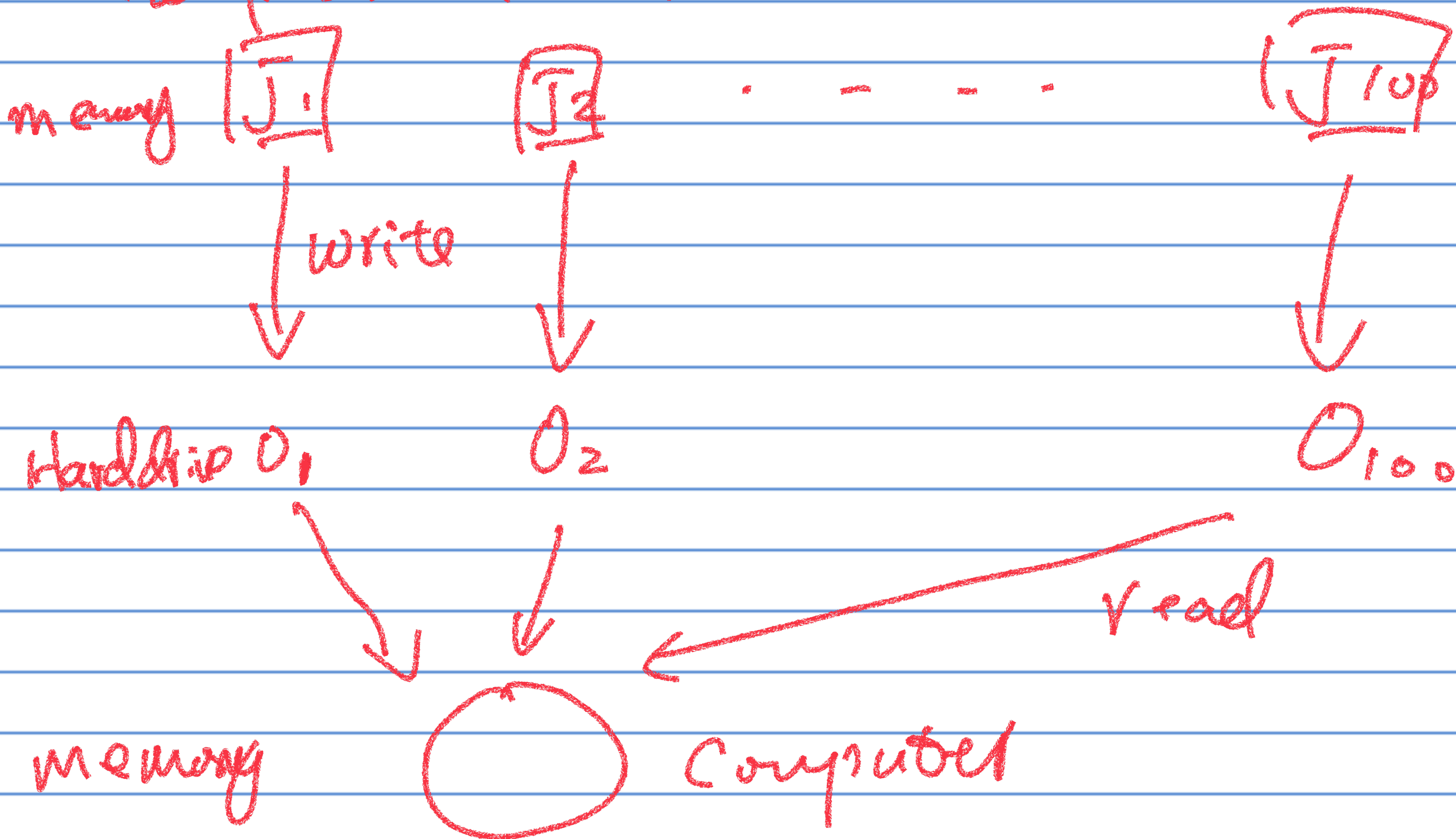


$L[[a]]$

$L[a]:$



Computer cluster



find max

$x[i]$

$x = [3$

4

7

8

9

1

$2]$

$i = 3$

$j.m$

3

$\rightarrow 4$

$\rightarrow 7$

$\rightarrow 8$

$\rightarrow 9$

$i(x.m)$

1

$\rightarrow 2$

$\rightarrow 3$

$\rightarrow 4$

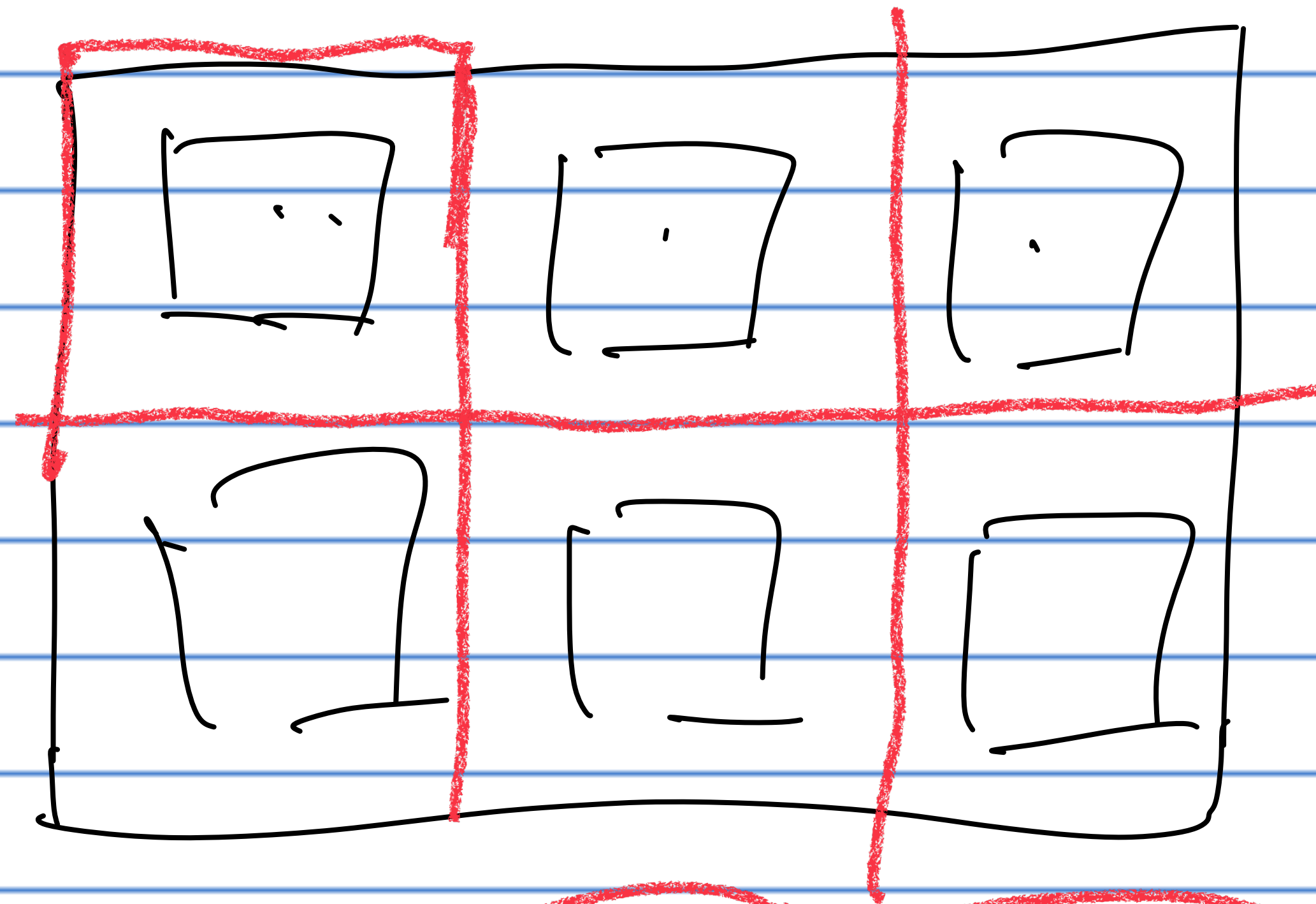
$\rightarrow 5$

$\Rightarrow$

9

$\leftarrow$

2:16



$m_{\text{row}} = c(2, 3)$

$m_{\text{col}} = c(2, 3)$

$xlim, ylim$

$mar = c(1, 2, 3, 4)$

$= c(2, 2, 3, 0)$

$x \leftarrow 1$

